> 🛠️ **2026.10 수정본 (v11)** — 원본: ch02_01_01B_knn_code_pratice.ipynb
>
> 타이타닉 `train.csv` 를 저장소의 `data/titanic/train.csv` 에서 읽도록 경로만 고쳤어요. 코랩에서는 이 파일을 같은 경로로 올려 주세요.

ch2 지도학습 - knn - 실습

- Machine Learning with sklearn @ DJ,Lim
- date : 21/10

실습해 보기

- titanic 데이터 셋을 활용하여 knn 모델을 구현한다.
- 가장 높은 일반화 성능을 갖는 k의 값은 무엇인지 찾아보자.

01 데이터 준비

02 머신러닝 모델 만들고 예측하기

01 데이터 준비

In [1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
import numpy as np
```

In [2]:
```
dat = pd.read_csv("data/titanic/train.csv")  # 2026.10 수정: 데이터 폴더 경로
dat
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376

891 rows × 12 columns

In [3]:
```
dat.columns
```

Out[3]:
```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='str')
```

In [4]:
```
dat.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

데이터 선택 및 나누기

In [5]:
```
X = dat[ ['Pclass' , 'SibSp'] ]
y = dat['Survived']

# 90% : 학습용, 10% : 테스트용
X_train, X_test, y_train, y_test = train_test_split(X, y,
                                                    test_size=0.1,
                                                    random_state=0)

X_train.shape, X_test.shape, y_train.shape, y_test.shape
```

Out[5]:
```
((801, 2), (90, 2), (801,), (90,))
```

In [6]:
```
from sklearn.neighbors import KNeighborsClassifier
model = KNeighborsClassifier(n_neighbors=2)
model.fit(X_train, y_train)

### 예측시키기
pred = model.predict(X_test)
```

In [7]:
```
(pred == y_test).sum() / len(pred)
```

Out[7]:
```
np.float64(0.5666666666666667)
```

In [8]:
```
print("테스트 세트의 정확도 : {:.2f}".format(np.mean(pred == y_test)))
```

```
테스트 세트의 정확도 : 0.57
```

결측치 처리 및 레이블 인코딩

In [9]:
```
dat.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

In [10]:
```
dat.head()
```

Out[10]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	7
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	5
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	

In [11]:
```
mapping = { "male":1, 'female':2 }
dat['Sex_num'] = dat['Sex'].map(mapping)
dat.head()
```

Out[11]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	7
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	5
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	

In [12]:
```
val_mean = dat['Age'].mean()
dat['Age'] = dat['Age'].fillna( val_mean )
dat.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 13 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          891 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
 12  Sex_num      891 non-null    int64
dtypes: float64(2), int64(6), str(5)
memory usage: 90.6 KB
```

학습, 테스트 데이터 셋 나누기

In [13]:
```
X = dat[ ['Pclass' , 'SibSp', 'Sex_num', 'Age'] ]
y = dat['Survived']

# 90% : 학습용, 10% : 테스트용
X_train, X_test, y_train, y_test = train_test_split(X, y,
                                                    test_size=0.1,
                                                    random_state=0)

X_train.shape, X_test.shape, y_train.shape, y_test.shape
```

Out[13]:
```
((801, 4), (90, 4), (801,), (90,))
```

In [14]:
```
from sklearn.neighbors import KNeighborsClassifier
model = KNeighborsClassifier(n_neighbors=2)
model.fit(X_train, y_train)
### 예측시키기
pred = model.predict(X_test)
print("테스트 세트의 정확도 : {:.2f}".format(np.mean(pred == y_test)))
```

```
테스트 세트의 정확도 : 0.73
```

k값에 따른 정확도 확인

In [15]:
```
tr_acc = []
test_acc = []
k_nums = range(1, 22, 2)# 1,3,5~21

for n in k_nums:
    # 모델 선택 및 학습
    model = KNeighborsClassifier(n_neighbors=n)
    model.fit(X_train, y_train)

    # 정확도 구하기
    acc_tr = model.score(X_train, y_train)
    acc_test = model.score(X_test, y_test)

    # 정확도 값 저장.
    tr_acc.append(acc_tr)
    test_acc.append(acc_test)

    print("k : ", n)
    print("학습용셋 정확도 {:.3f}".format(acc_tr) )
    print("테스트용셋 정확도 {:.3f}".format(acc_test) )
```

```
k :  1
학습용셋 정확도 0.881
테스트용셋 정확도 0.767
k :  3
학습용셋 정확도 0.859
테스트용셋 정확도 0.744
k :  5
학습용셋 정확도 0.836
테스트용셋 정확도 0.733
k :  7
학습용셋 정확도 0.828
테스트용셋 정확도 0.733
k :  9
학습용셋 정확도 0.829
테스트용셋 정확도 0.711
k :  11
학습용셋 정확도 0.825
테스트용셋 정확도 0.744
k :  13
학습용셋 정확도 0.820
테스트용셋 정확도 0.767
k :  15
학습용셋 정확도 0.804
테스트용셋 정확도 0.756
k :  17
학습용셋 정확도 0.799
테스트용셋 정확도 0.756
k :  19
학습용셋 정확도 0.785
테스트용셋 정확도 0.722
k :  21
학습용셋 정확도 0.783
테스트용셋 정확도 0.744
```

In [16]:
```
import seaborn as sns
print(sns.__version__)
```

```
0.13.2
```

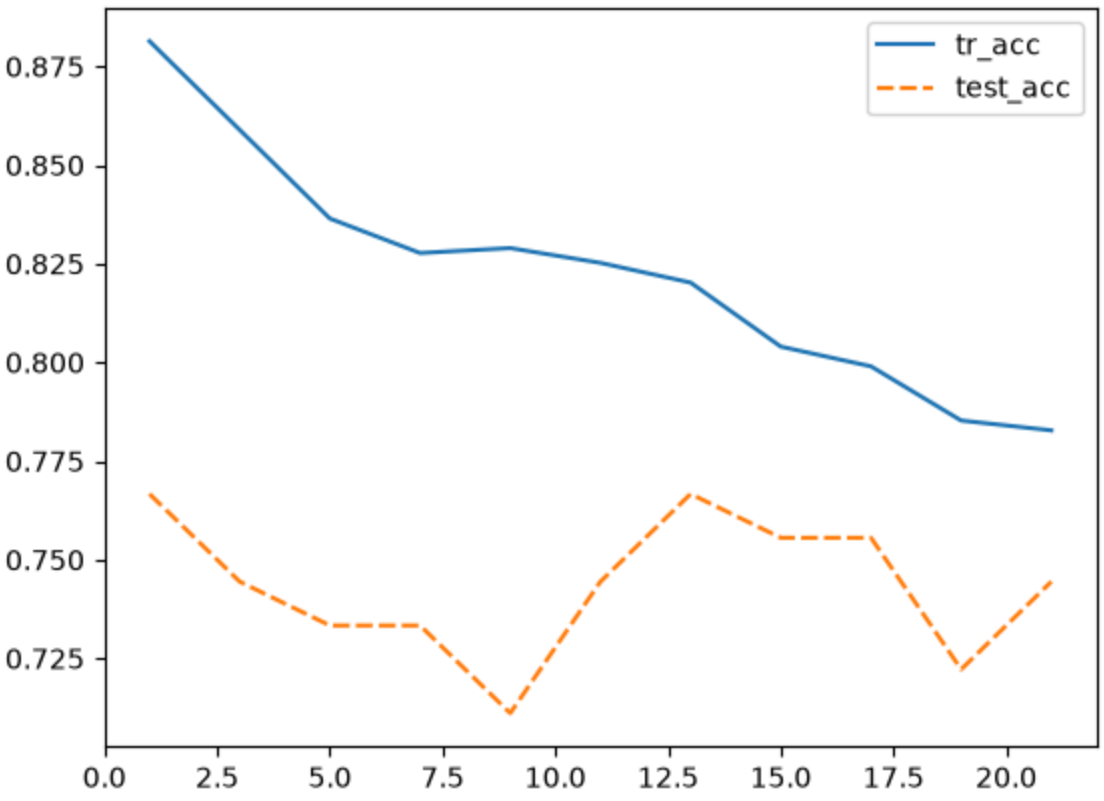
In [17]:
```
# tr_acc = []
# test_acc = []
dat = { "tr_acc":tr_acc, "test_acc":test_acc }
data_df = pd.DataFrame(dat, index=range(1, 22, 2))
data_df
```

Out[17]:

	tr_acc	test_acc
1	0.881398	0.766667
3	0.858926	0.744444
5	0.836454	0.733333
7	0.827715	0.733333
9	0.828964	0.711111
11	0.825218	0.744444
13	0.820225	0.766667
15	0.803995	0.755556
17	0.799001	0.755556
19	0.785268	0.722222
21	0.782772	0.744444

In [18]:
```
import matplotlib.pyplot as plt
```

In [19]:
```
sns.lineplot(data=data_df, palette="tab10")
plt.show()
```



In [ ]: